Massachusetts Institute of Technology

Single Event Views of the GEMs

I looked, and there before me was a pale horse! Its rider was named Death, …given power over a fourth of the earth to kill by sword, famine and plague, and by the wild beasts of the earth.

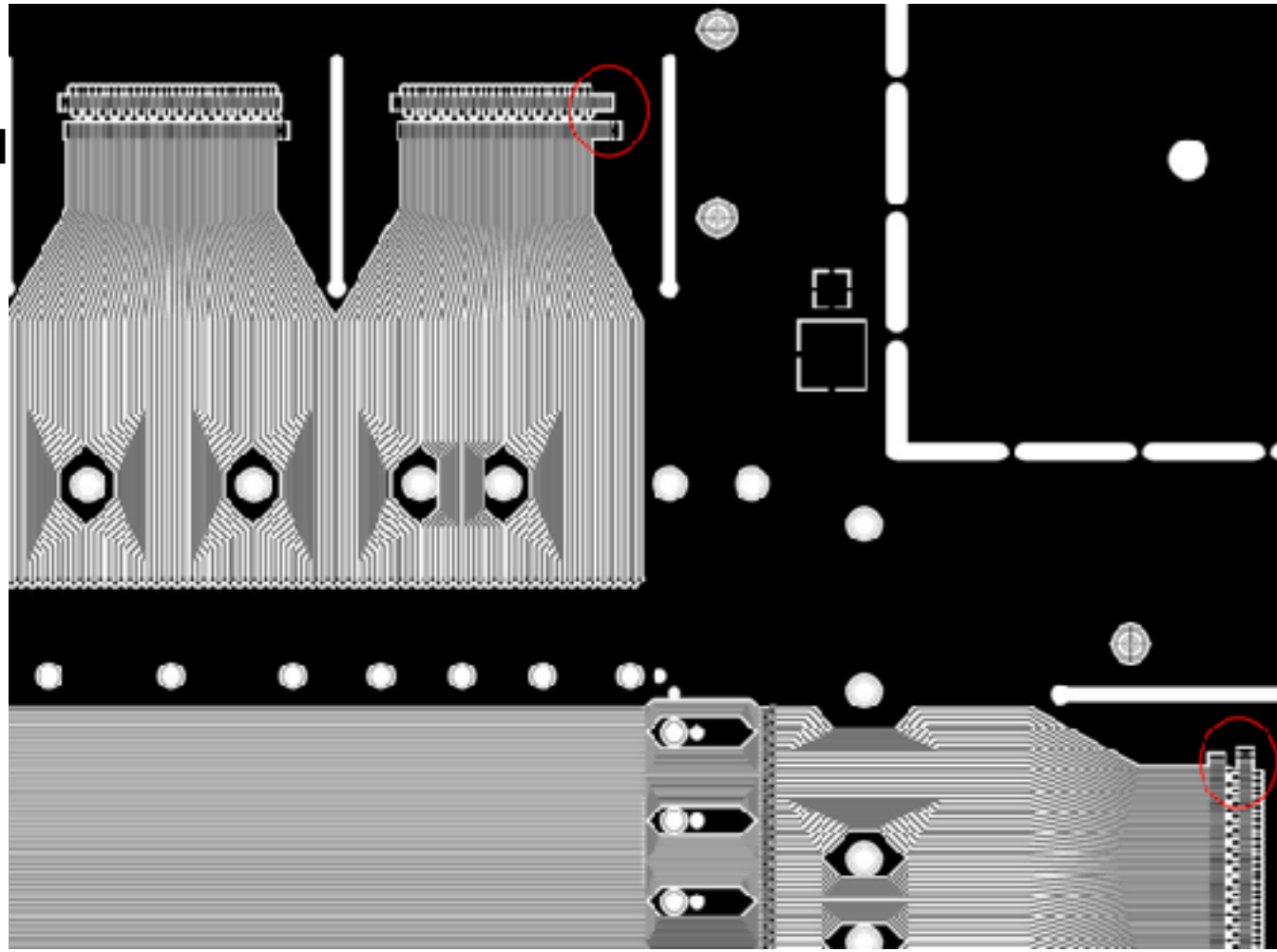
Mahmoud and Ethan

April 7, 2026

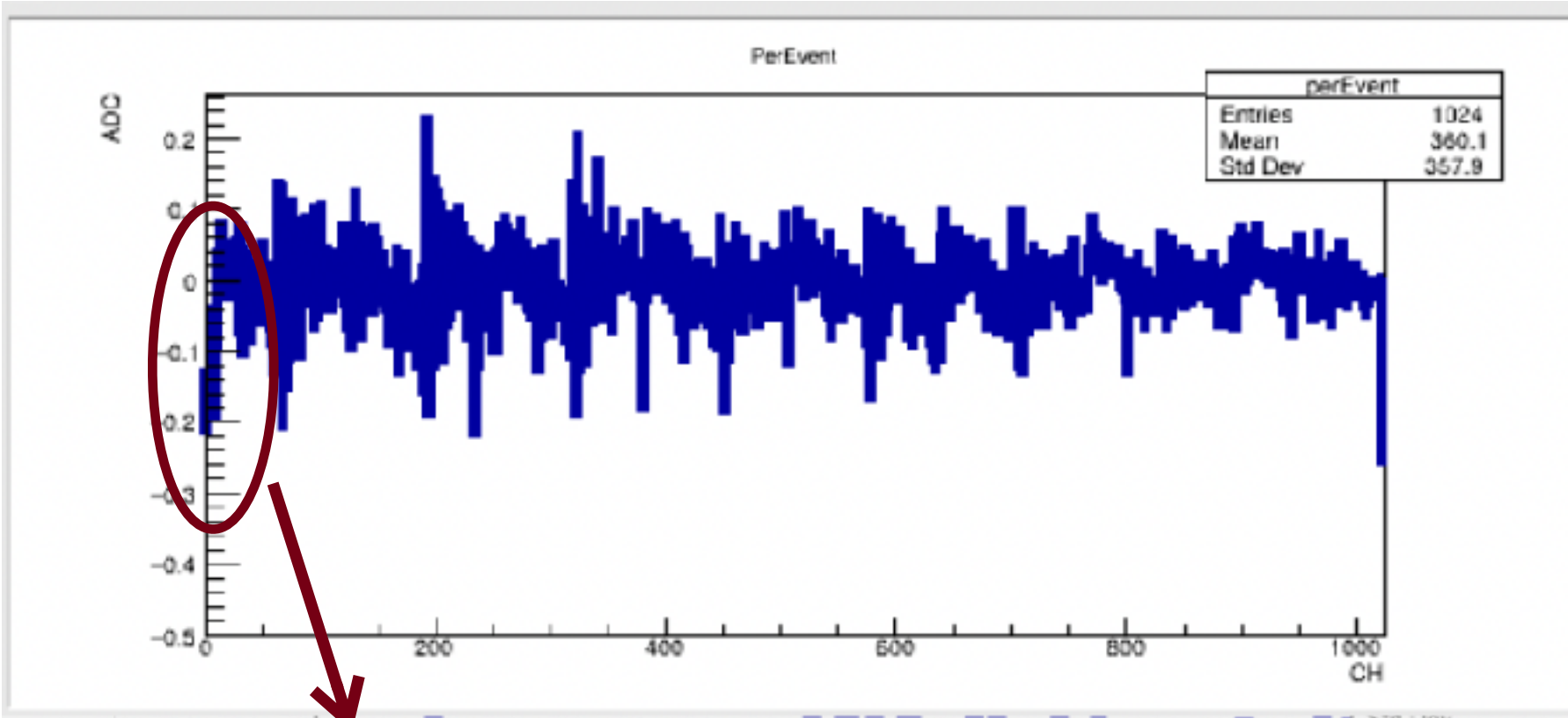
GEM Channel Numbering
A nightmare

- Recall from Dulitha's presentation in January, we have APV strips not connected to readout
- 11 on each end of long side and 6 on each end of short side
- Not major impact but these contribute to common mode subtraction calculation
    - Can cause spurious hits

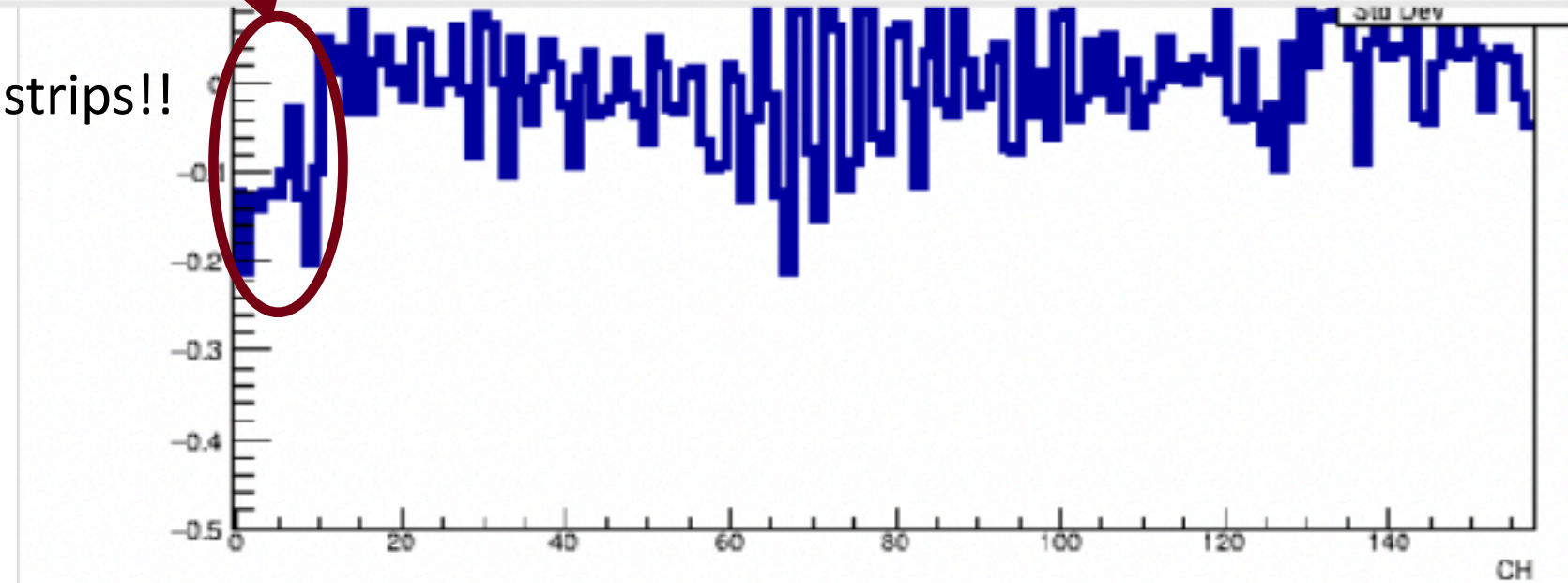
- FIX NOT IMPLEMENTED ANYWHERE AFAIK



Not connected channels



Not connected strips!!



Fix in .ini files?

```
apv:0=0,128,0,"APV0"
apv:1=128,128,0,"APV1"
apv:2=256,128,0,"APV2"
apv:3=384,128,0,"APV3"
apv:4=512,128,0,"APV4"
apv:5=640,128,0,"APV5"
apv:6=768,128,0,"APV6"
apv:7=896,128,0,"APV7"
apv:8=1024,128,0,"APV0"
apv:9=1152,128,0,"APV1"
apv:10=1280,128,0,"APV2"
apv:11=1408,128,0,"APV3"
apv:12=1536,128,0,"APV4"
```

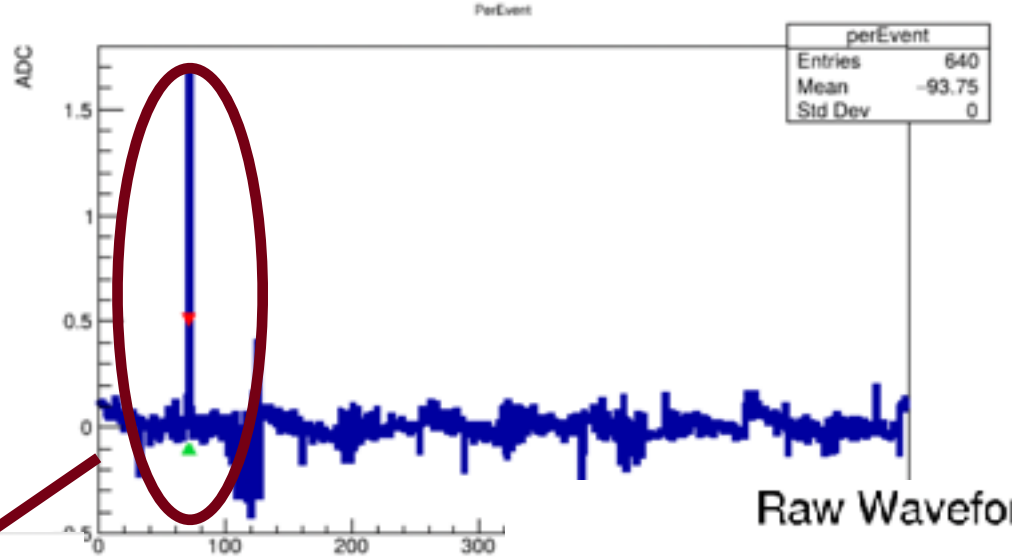
?????
1024 or
1013 or
1002?

```
apv:0=11,117,11,"APV0"
apv:1=128,128,0,"APV1"
apv:2=256,128,0,"APV2"
apv:3=384,128,0,"APV3"
apv:4=512,128,0,"APV4"
apv:5=640,128,0,"APV5"
apv:6=768,128,0,"APV6"
apv:7=896,117,0,"APV7"
apv:8=1024,128,0,"APV0"
apv:9=1152,128,0,"APV1"
apv:10=1280,128,0,"APV2"
apv:11=1408,128,0,"APV3"
apv:12=1536,128,0,"APV4"
```

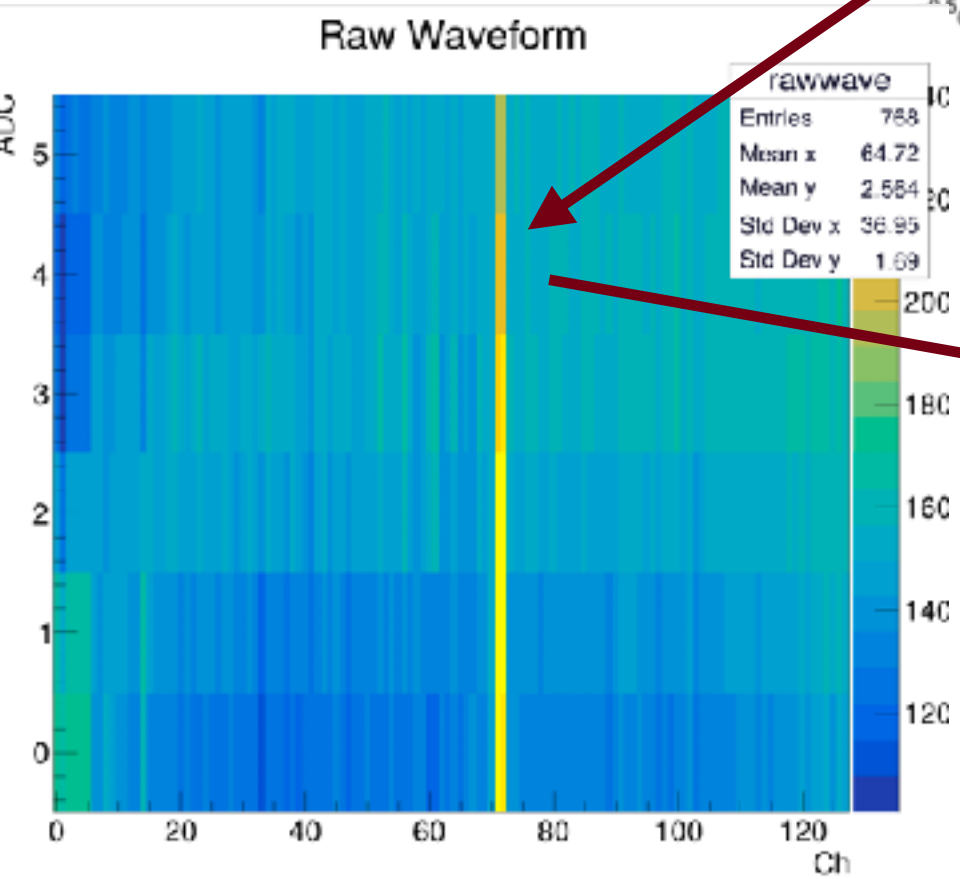
We read out 128 channels, we just don't want to plot some. NOT IMPLEMENTED!!!

```
apv:0=11,128,11,"APV0"
apv:1=128,128,0,"APV1"
apv:2=256,128,0,"APV2"
apv:3=384,128,0,"APV3"
apv:4=512,128,0,"APV4"
apv:5=640,128,0,"APV5"
apv:6=768,128,0,"APV6"
apv:7=896,117,0,"APV7"
apv:8=1024,128,6,"APV0"
apv:9=1152,128,0,"APV1"
apv:10=1280,128,0,"APV2"
apv:11=1408,128,0,"APV3"
apv:12=1536,122,0,"APV4"
```

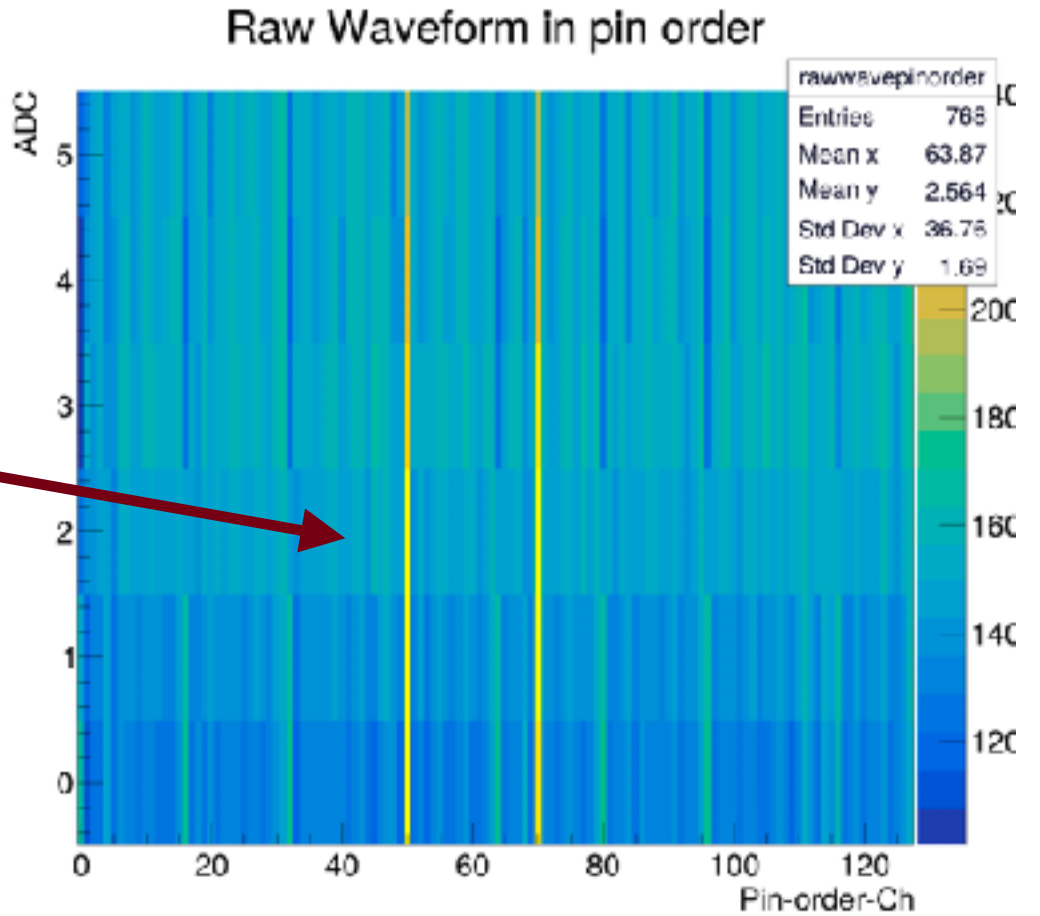
Single Events
Run 2796, cosmic



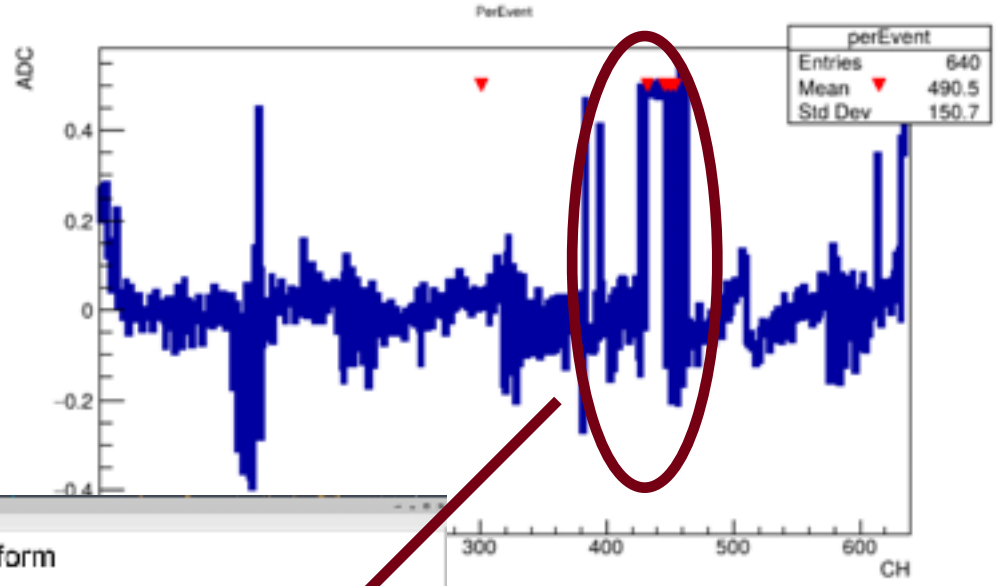
On GEM strips



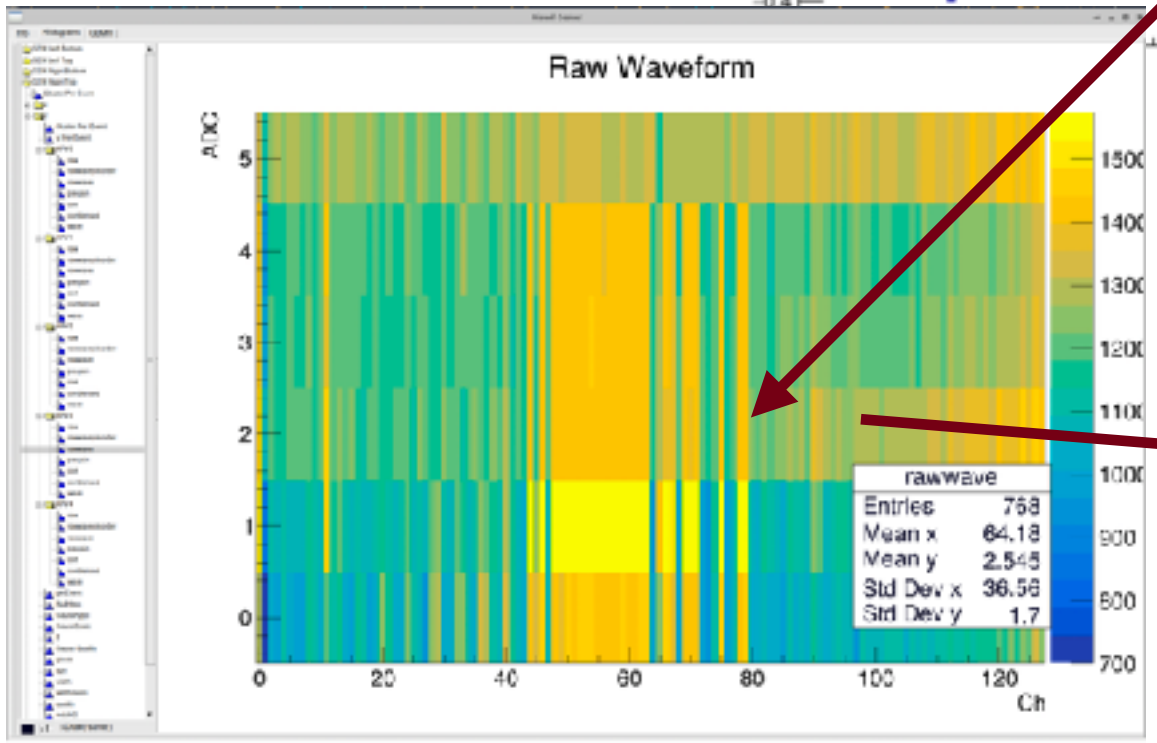
On APV channels



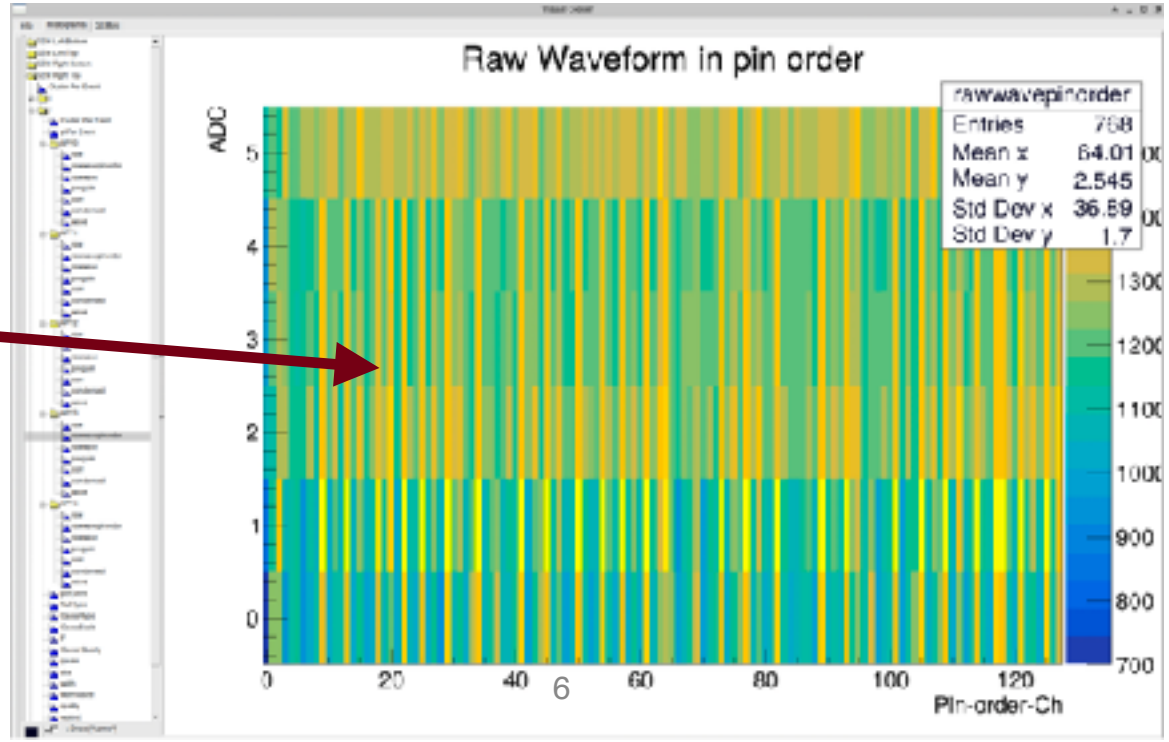
Single Events
Run 2796, cosmic



On GEM strips

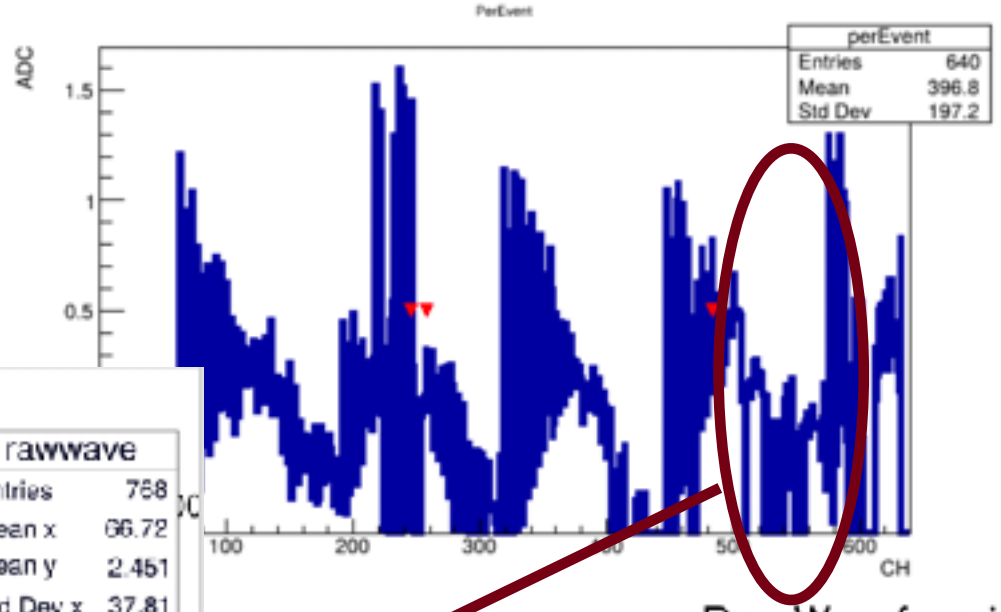


On APV channels

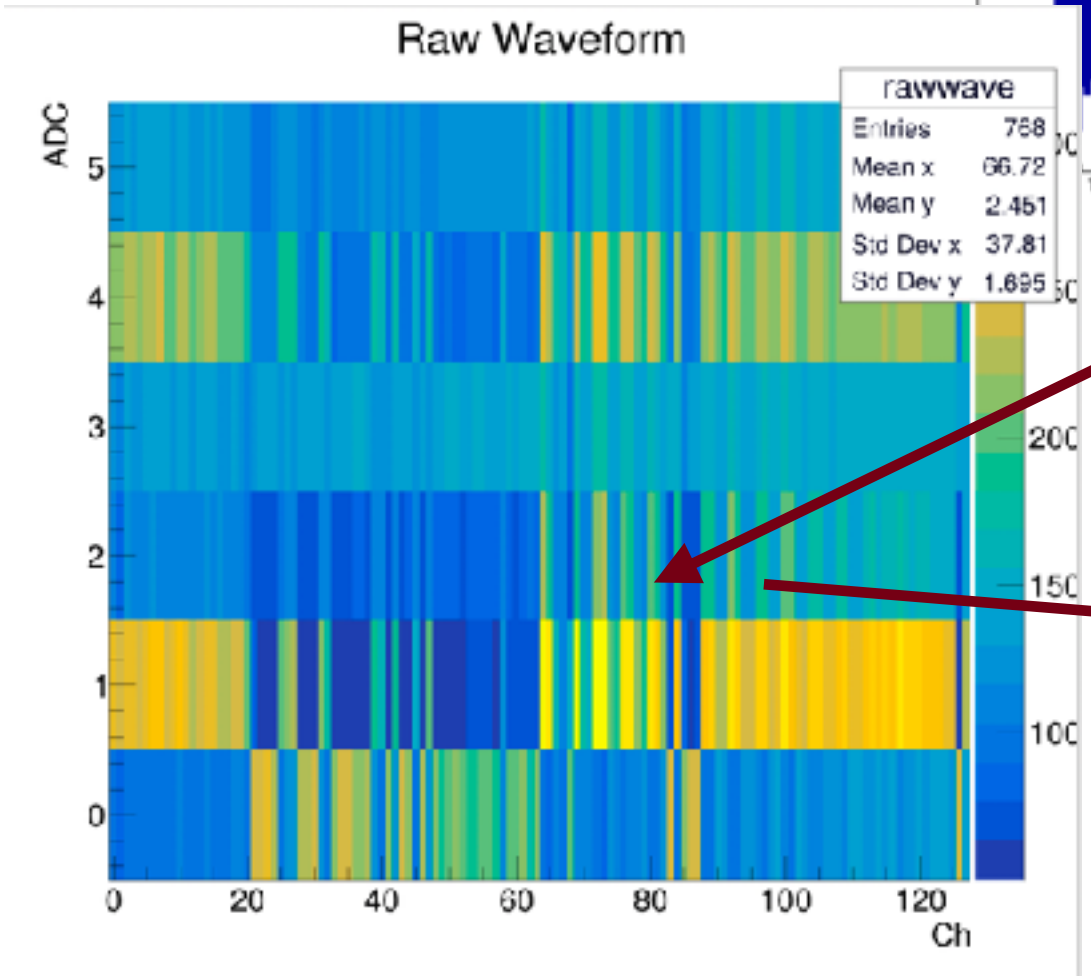


What is causing a large ADC readout on a series of GEM strips? A spark?

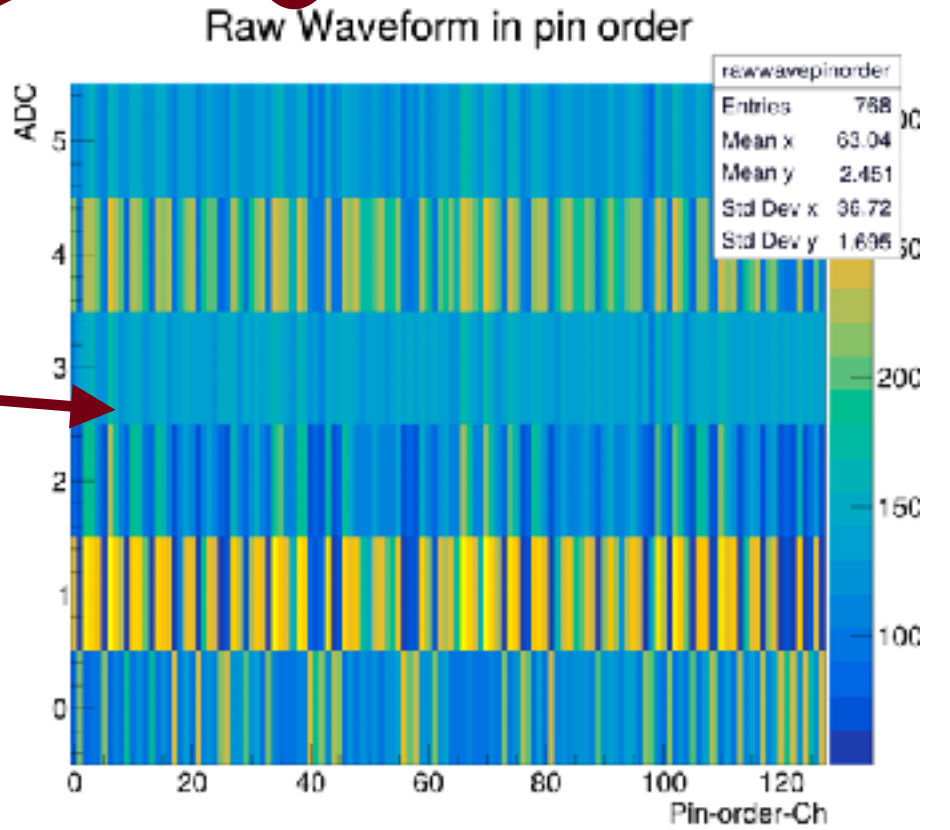
Single Events
Run 2796, cosmic



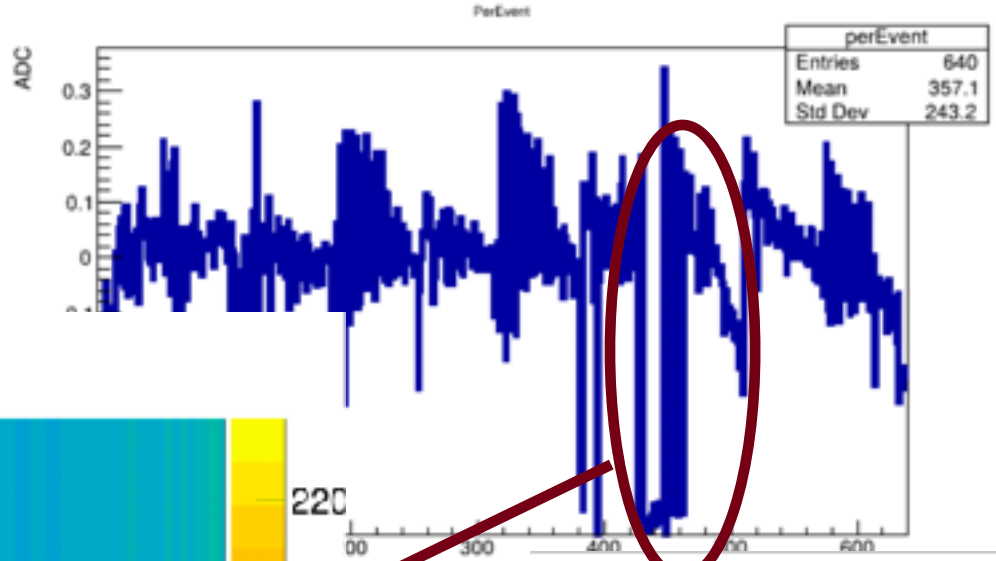
On GEM strips

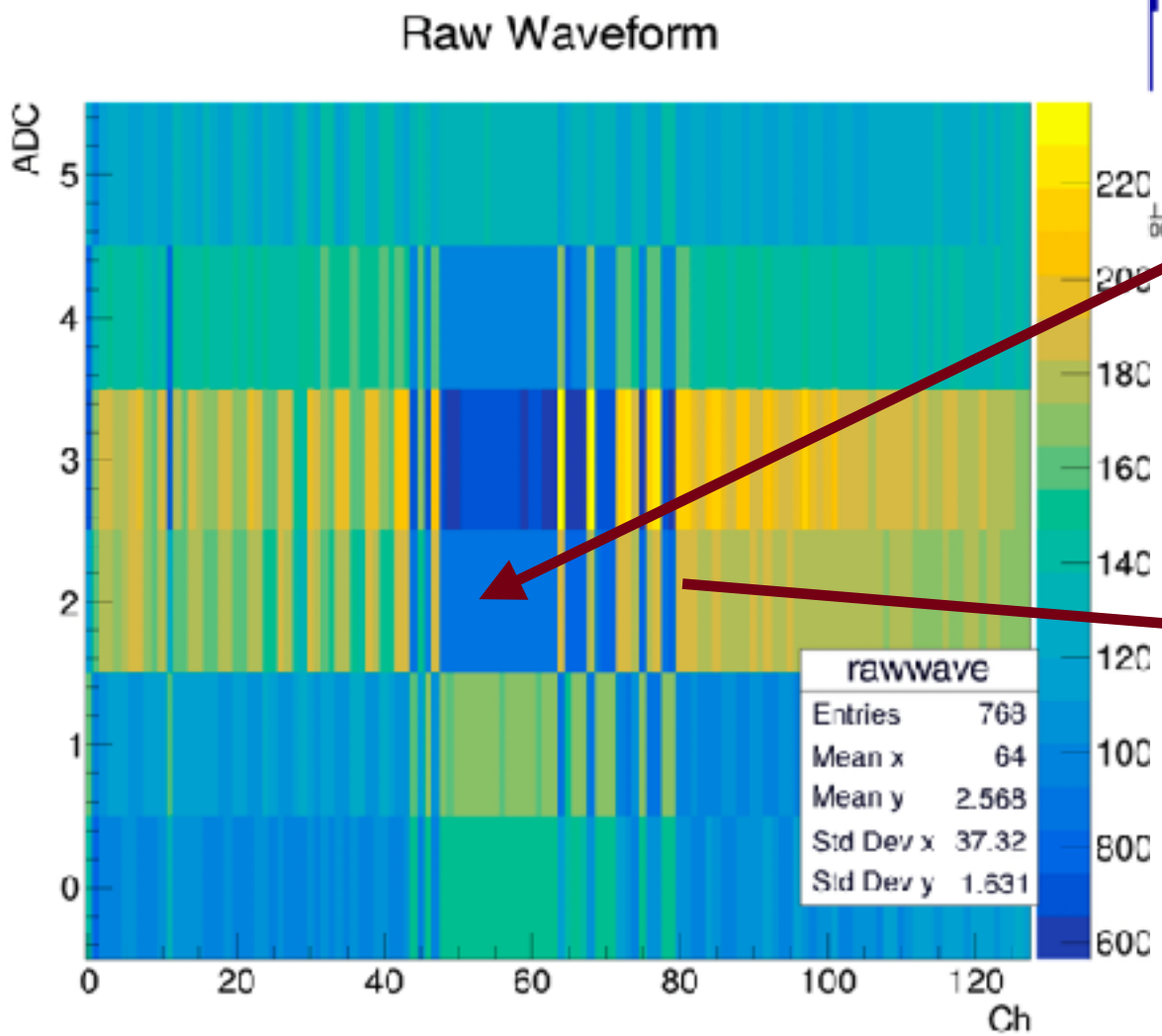


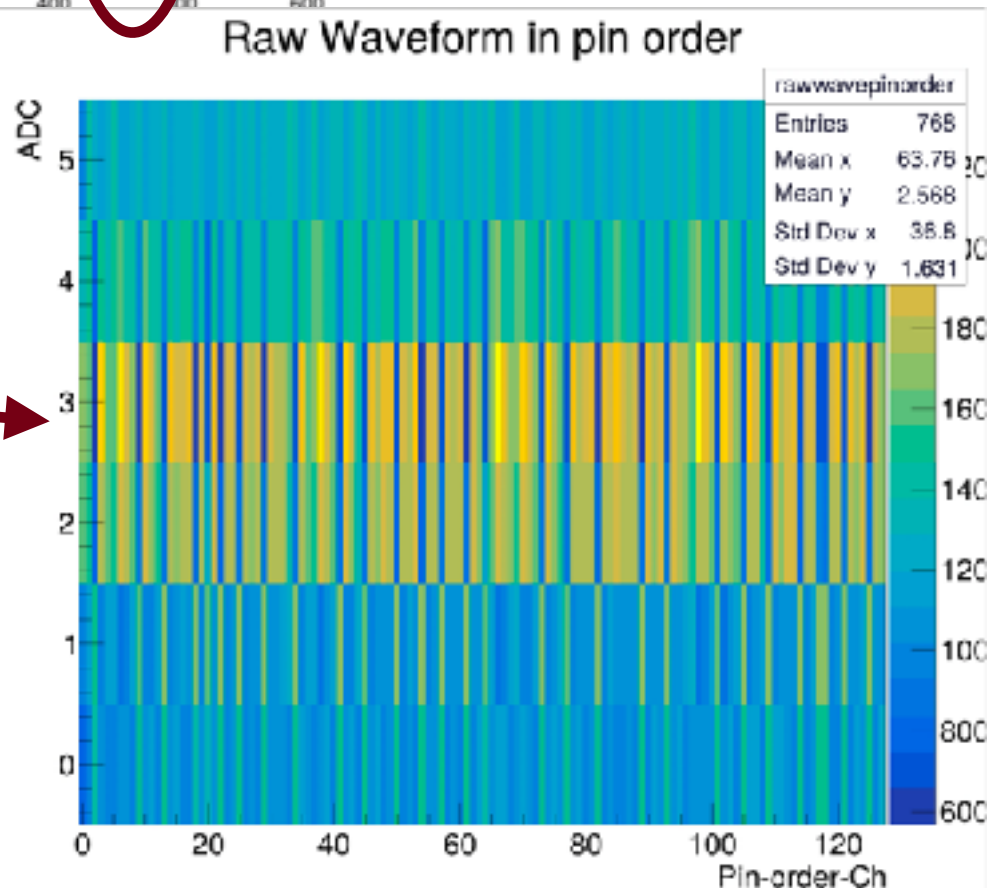
On APV channels



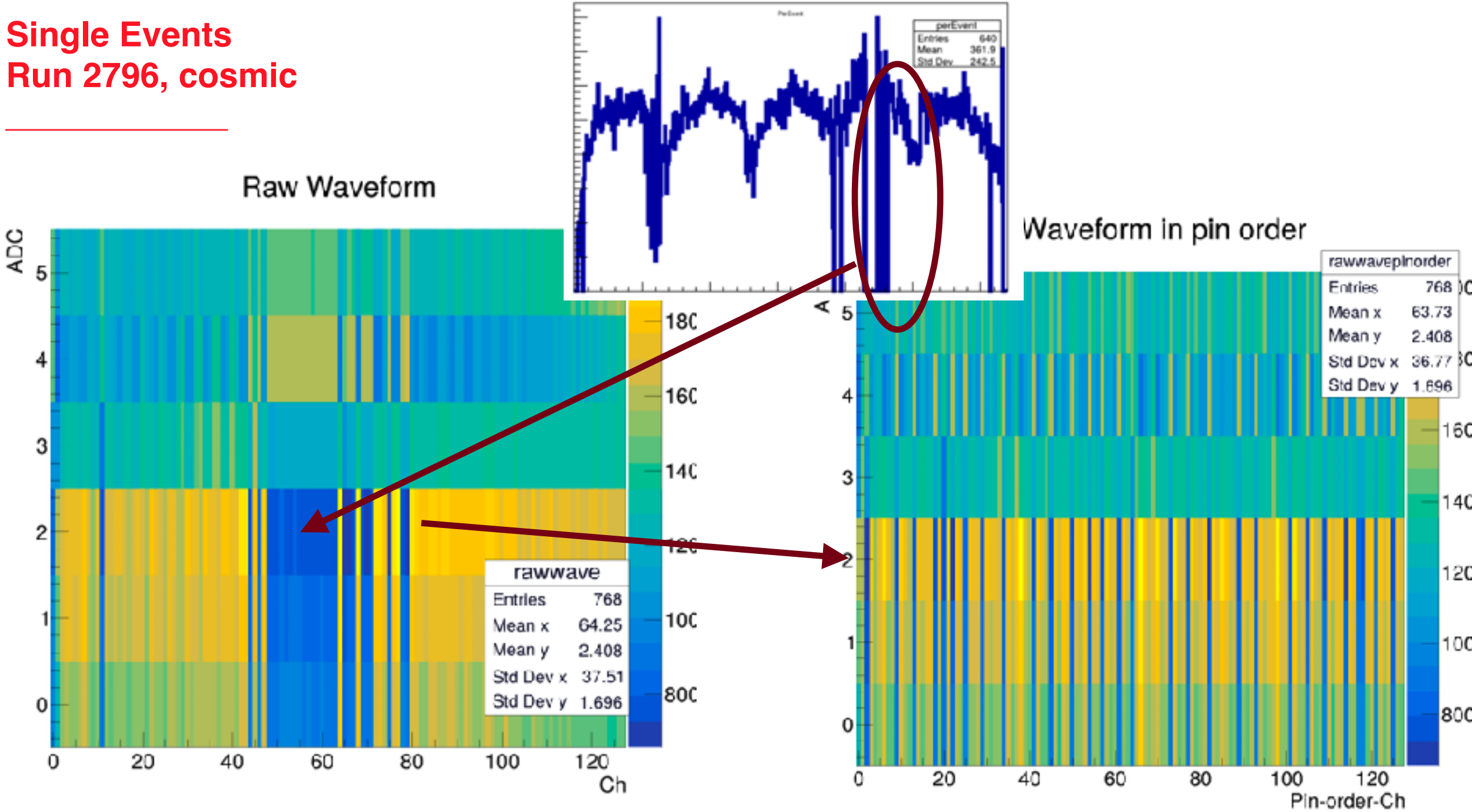
Single Events
Run 2796, cosmic



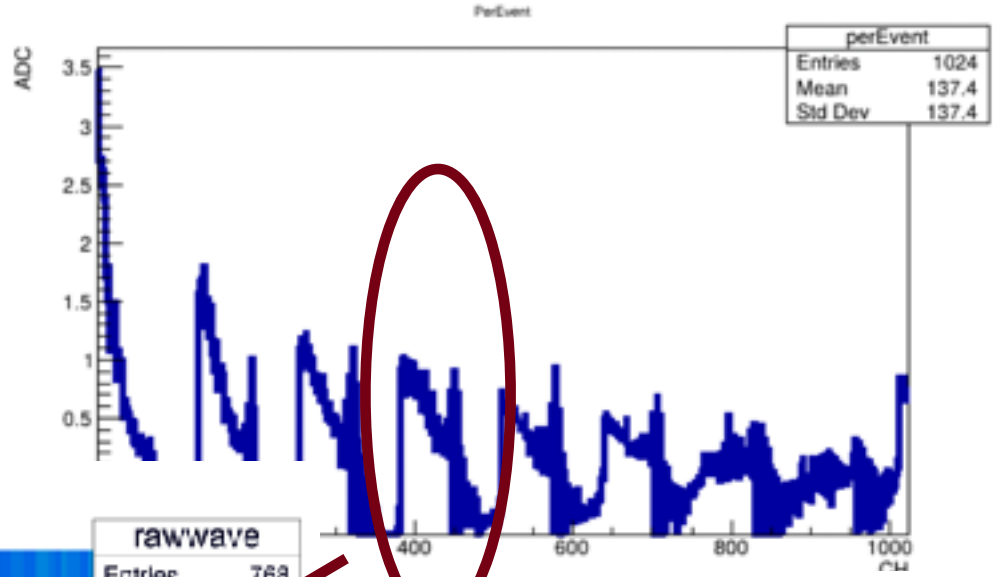


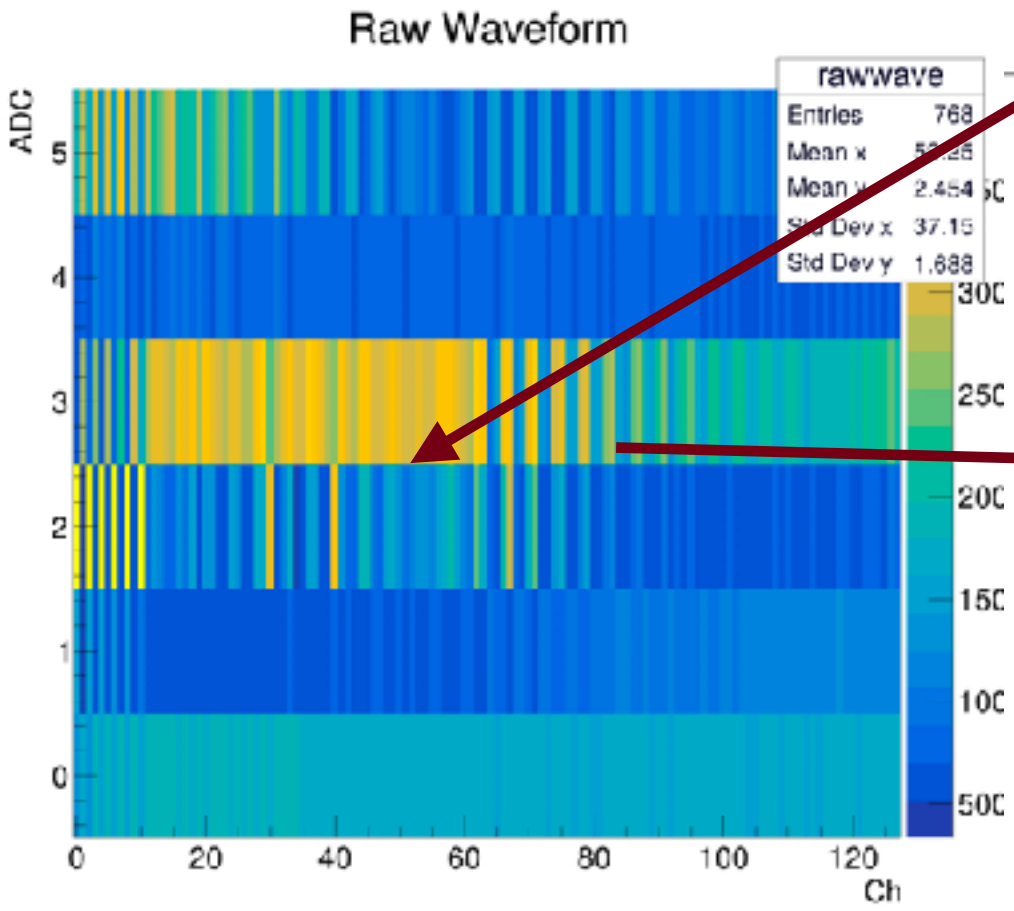


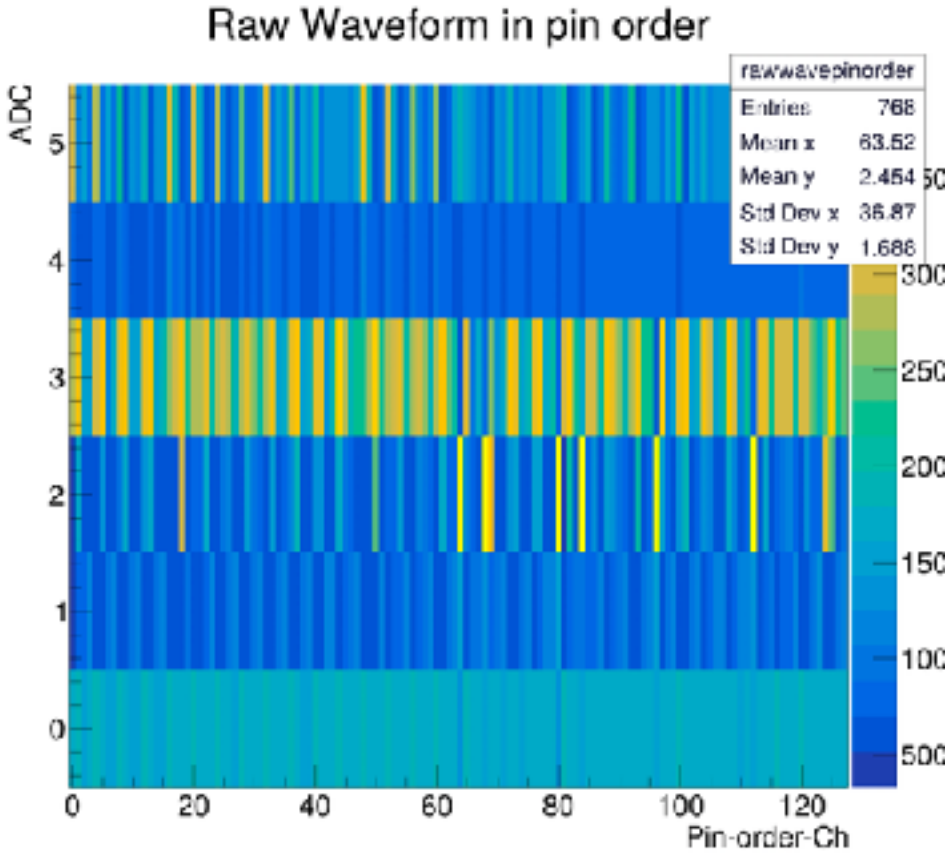
Single Events
Run 2796, cosmic



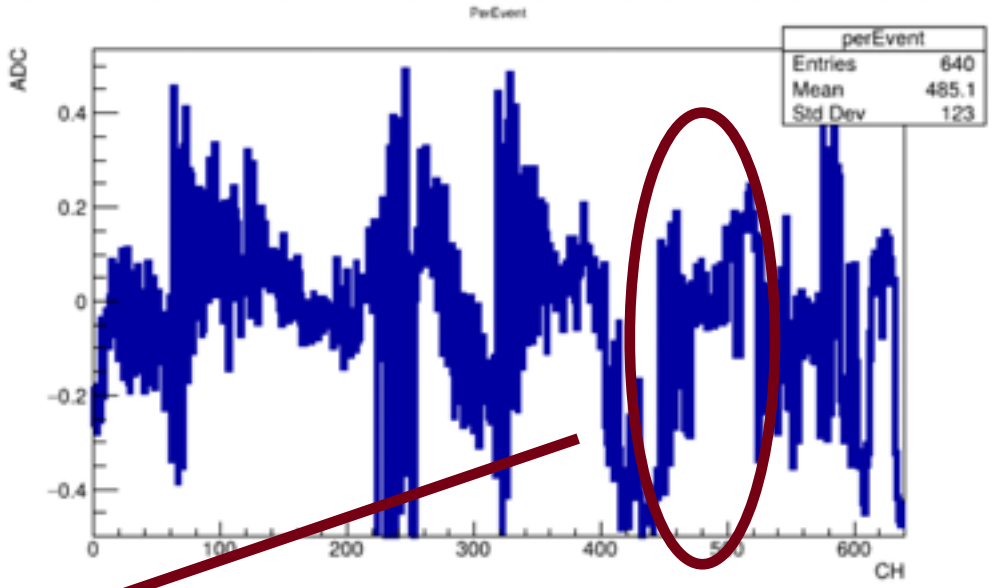
Single Events
Run 2796, cosmic

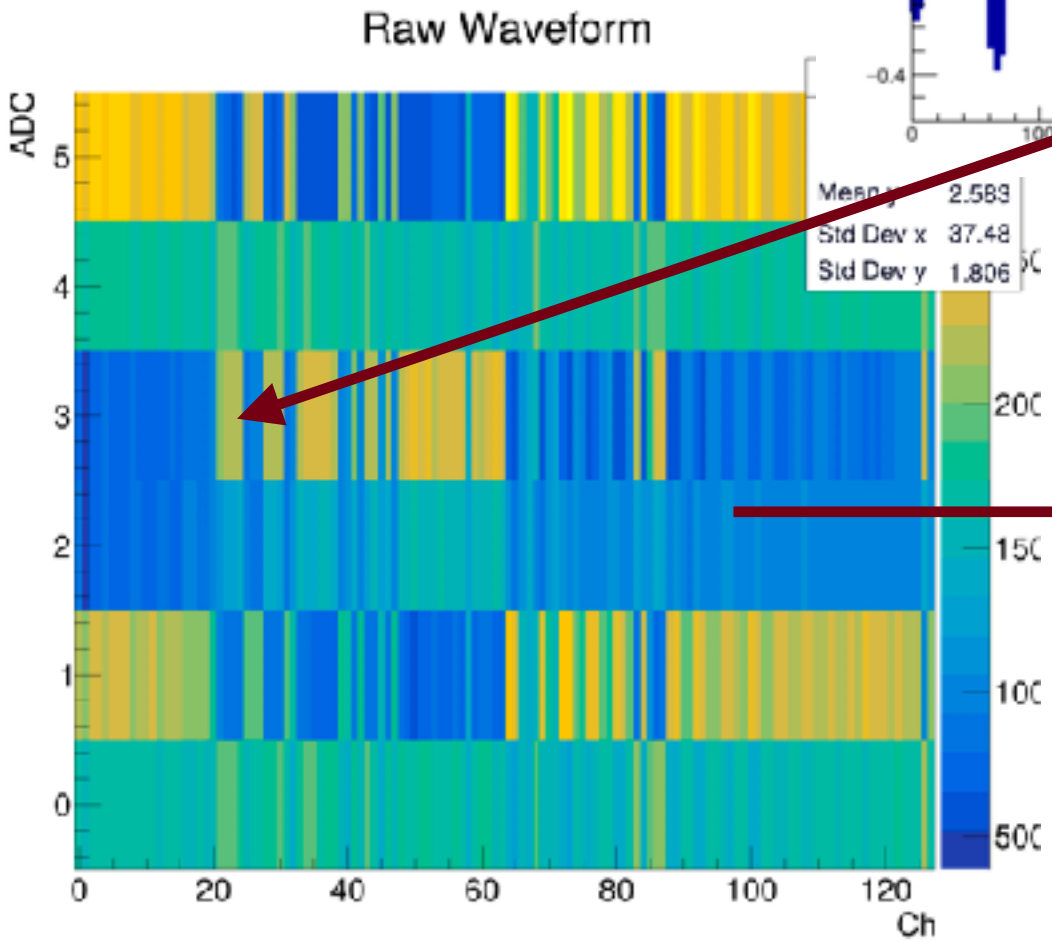


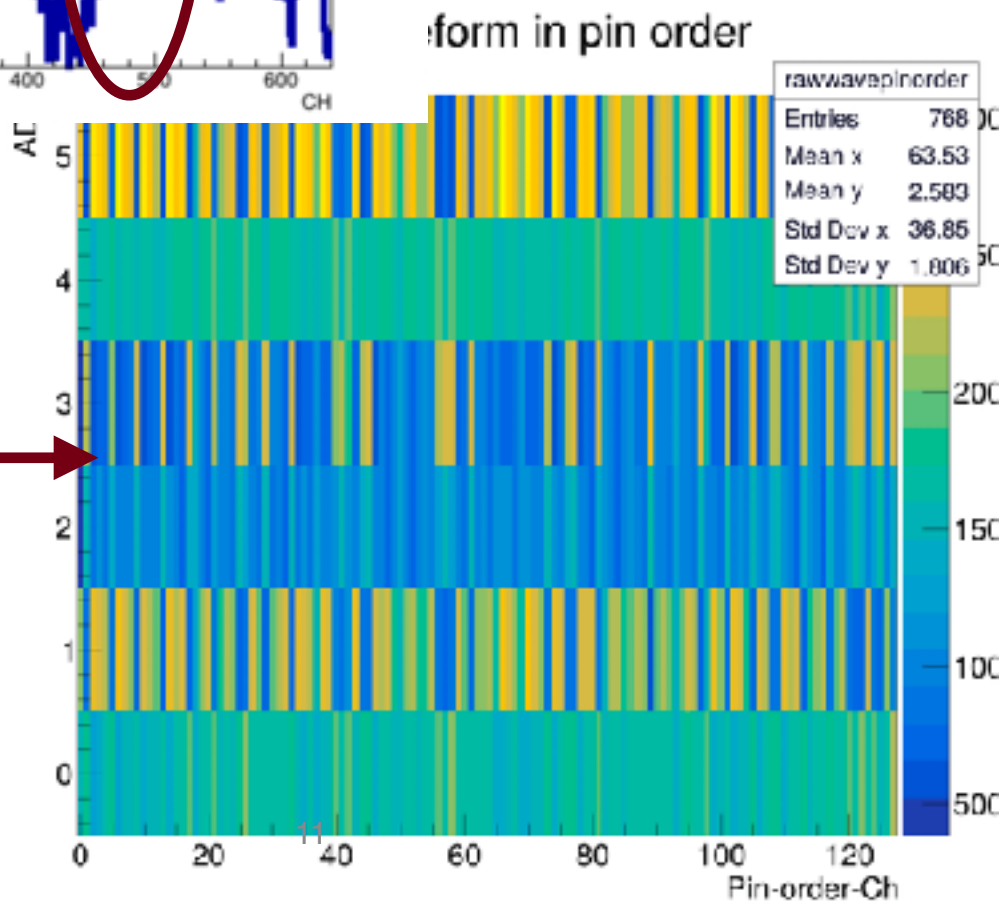




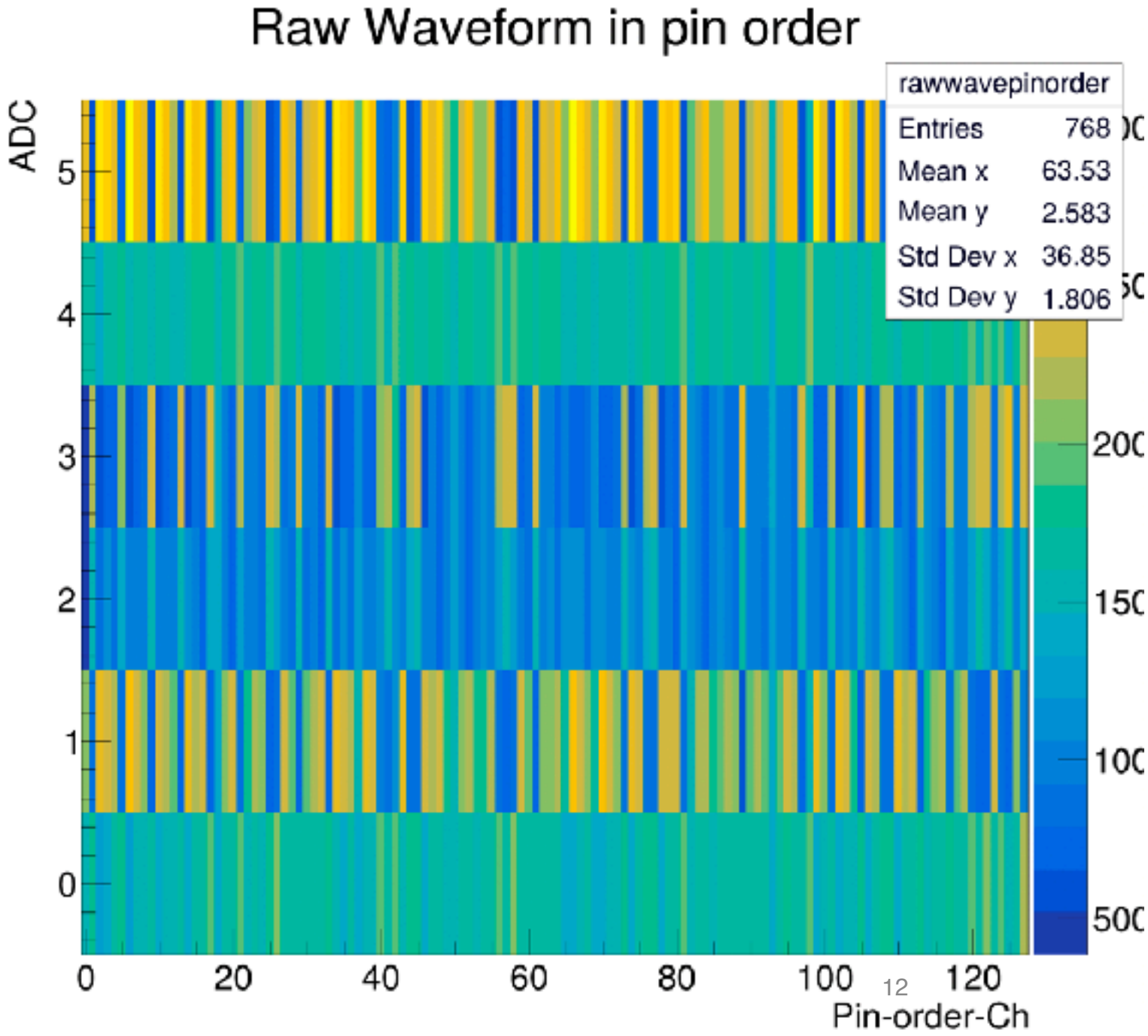
Single Events
Run 2796, cosmic



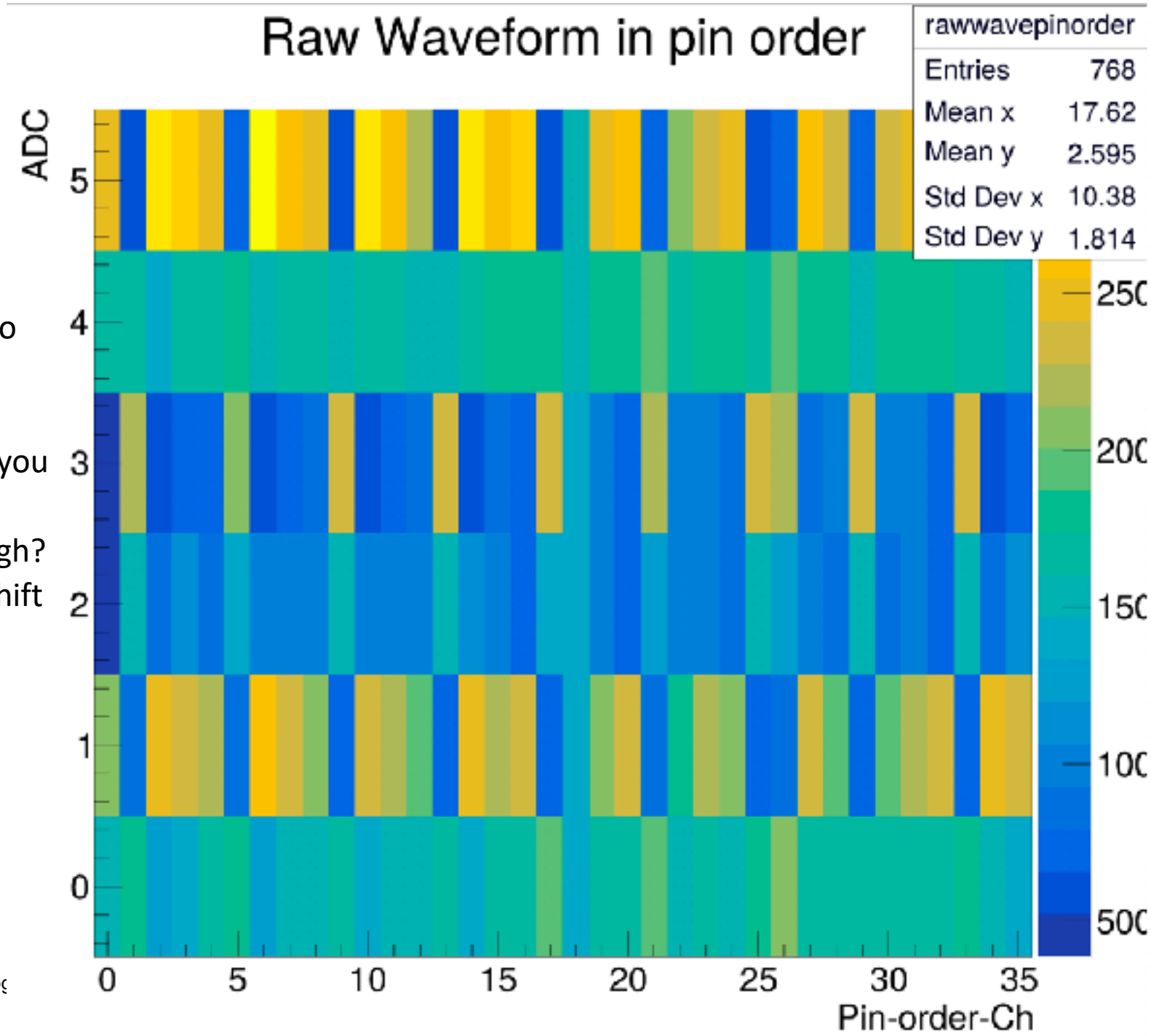




No reason for there to be a strip-level explanation. Likely comes from APV. Do you see a pattern here?
A series of 3 low-1 high? Left to right? And a shift of 2 in sample?



No reason for there to be a strip-level explanation. Likely comes from APV. Do you see a pattern here?
A series of 3 low-1 high? Left to right? And a shift of 2 in sample?



Next Steps

- **Fix missing channel issue**
- **Identify all different types of noise, seems to be several standard "behaviors", each may have different causes and solutions**
  - ***Roughly speaking*, seems that noise is correlated. As in, all APVs on all GEMs will look bad at the same time**
  - **Also, looks like non-dispersive generally worse than dispersive. Due to back scatter from the dump?**
- **Clean up clustering in gemini**

```
if (false)
{ // Previous Method
```

```
if (true)
{
```

Current gemini cluster finder selection method. Do not do this when writing code.

General Reminder
CutsDB

- **We have a cuts DB! Everyone should use it**
- **Makes it clear what cuts have been applied to what file**
- **Src/plugins/tools/cutsDB/init/cutsDB.ini**

In cutsDB.ini

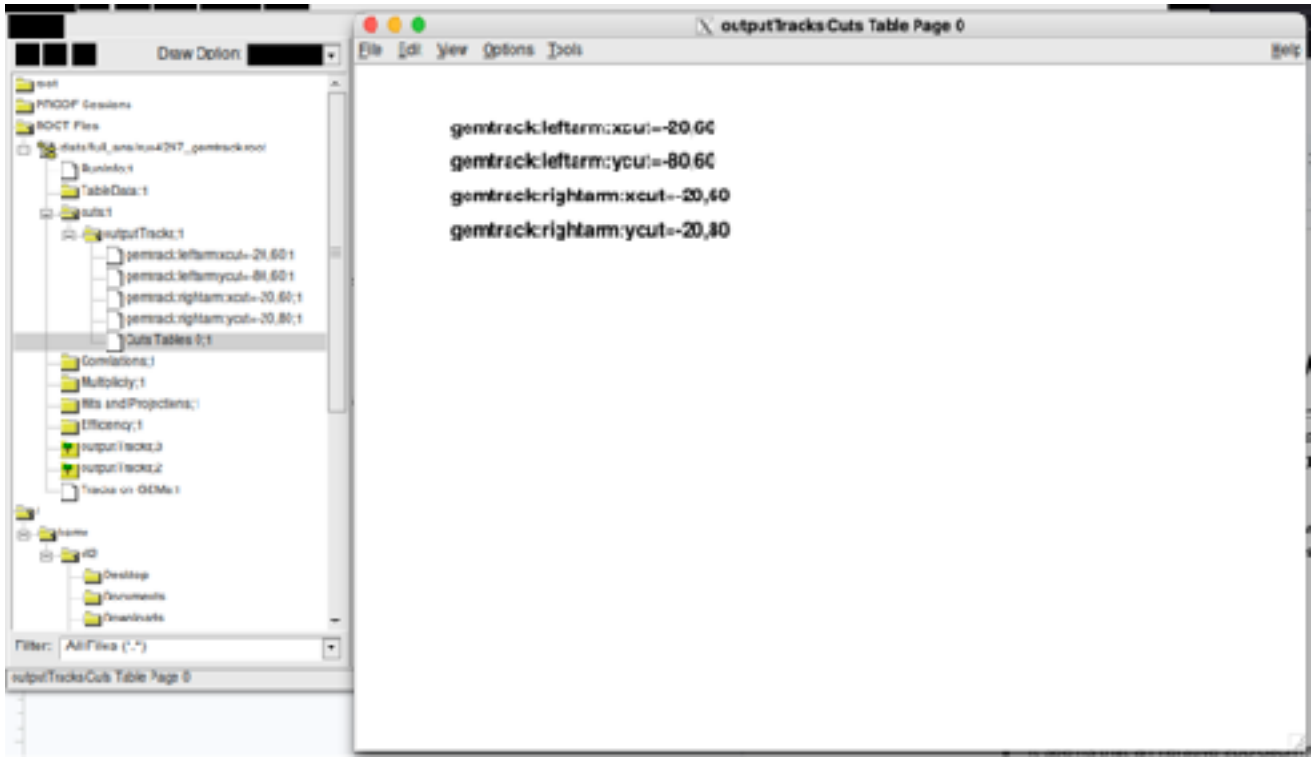
```
[run:2500]
    cut2d:"gemtrack:leftarm:xcut"=-20,60
    cut2d:"gemtrack:leftarm:ycut"=-80,60
    cut2d:"gemtrack:rightarm:xcut"=-20,60
    cut2d:"gemtrack:rightarm:ycut"=-20,80

[run:10000]
    cut2d:"gemtrack:leftarm:xcut"=-40,40
    cut2d:"gemtrack:leftarm:ycut"=-40,40
    cut2d:"gemtrack:rightarm:xcut"=-40,40
    cut2d:"gemtrack:rightarm:ycut"=-40,40
```

In GEMTrack.cpp

```
left_arm_cut_x = cutsDB->get("gemtrack:leftarm:xcut");
left_arm_cut_y = cutsDB->get("gemtrack:leftarm:ycut");
right_arm_cut_x = cutsDB->get("gemtrack:rightarm:xcut");
right_arm_cut_y = cutsDB->get("gemtrack:rightarm:ycut");
```

General Reminder
CutsDB

- **We have a cuts DB! Everyone should use it**
- **Makes it clear what cuts have been applied to what file**
- **Src/plugins/tools/cutsDB/init/cutsDB.ini**



In the output .root file